Experiment - 7
(Cryptography - Diffie-Hellman Algorithm)

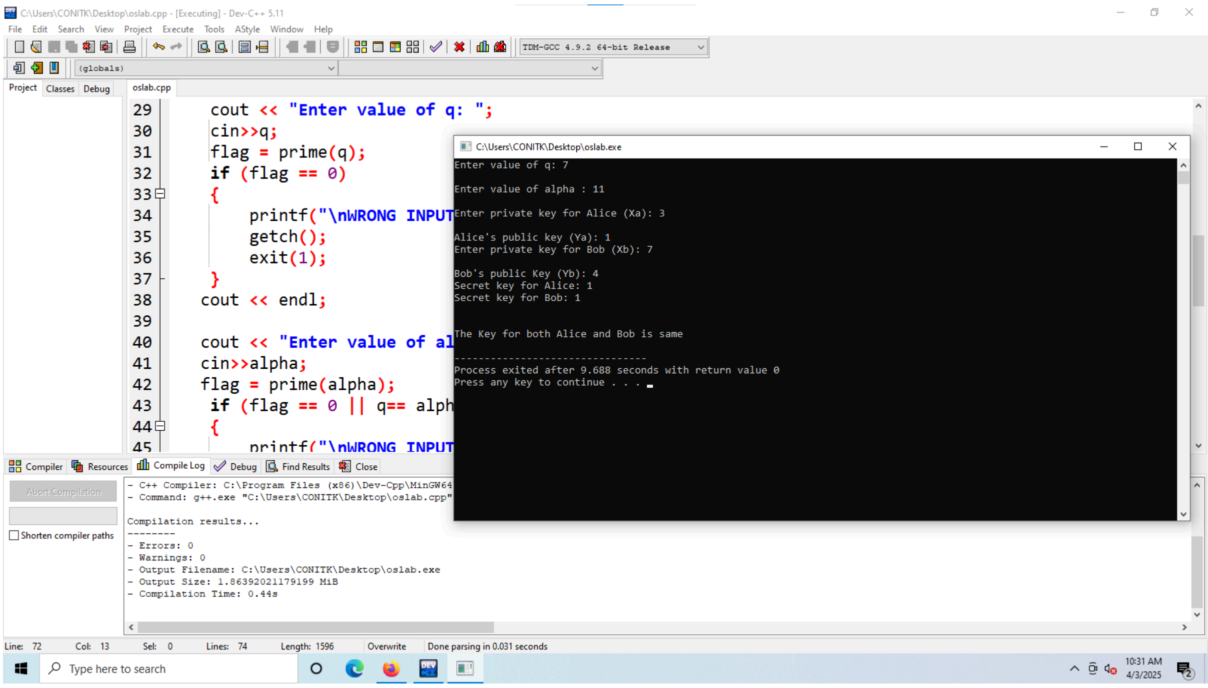
CODE :

```
#include <iostream>
#include <cmath>
#include<conio.h>

using namespace std;

long long int calculatePower(long long int base, long long int
exponent, long long int modulus) {
  if (exponent == 1)
      return base;
  else
      return (((long long int)pow(base, exponent)) % modulus);
}

int prime(long int pr)
{
      int i,j;
      j = sqrt(pr);
      for (i = 2; i <= j; i++)
      {
      if (pr % i == 0)
      return 0;
      }
      return 1;
}

int main() {
  long long int q, alpha, Xa,Xb, Ya, Yb, Ka, Kb;
```

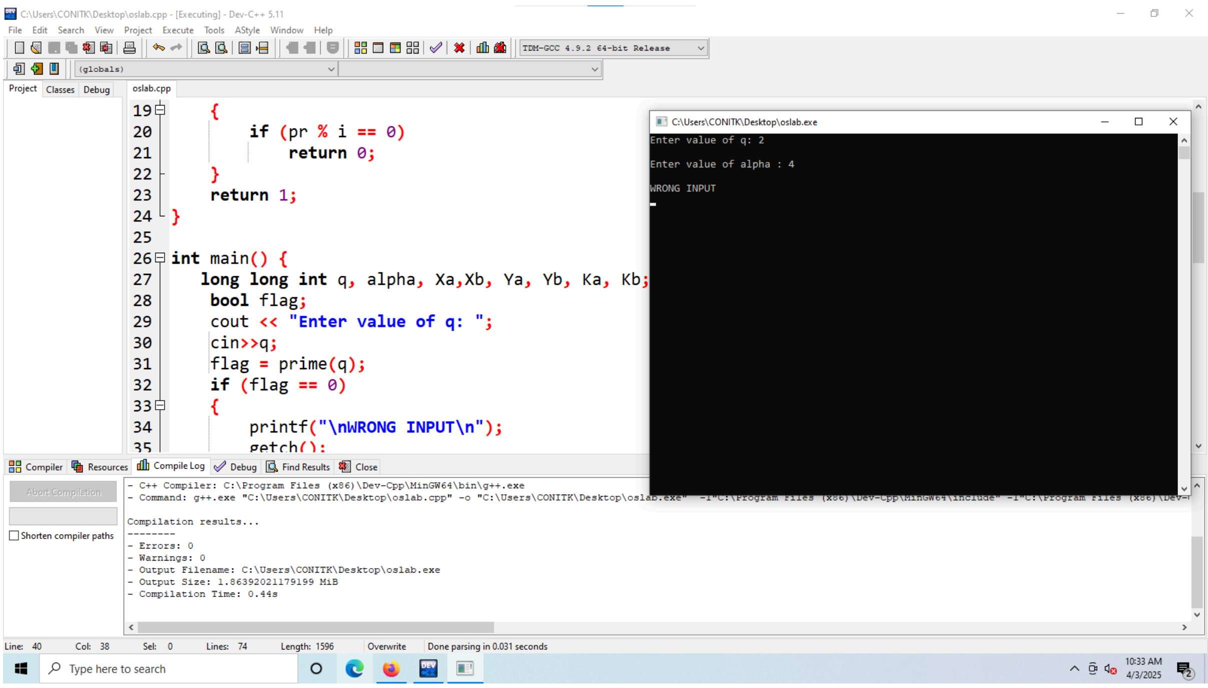
```
 bool flag;
     cout << "Enter value of q: ";
      cin>>q;
      flag = prime(q);
     if (flag == 0)
     {
     printf("\nWRONG INPUT\n");
     getch();
     exit(1);
     }
cout << endl;

cout << "Enter value of alpha : ";
cin>>alpha;
flag = prime(alpha);
     if (flag == 0 || q== alpha)
     {
     printf("\nWRONG INPUT\n");
     getch();
     exit(1);
     }
cout << endl;

cout << "Enter private key for Alice (Xa): ";
cin>>Xa;
cout<< endl;

Ya = calculatePower(alpha, Xa, q);
cout << "Alice's public key (Ya): "<<Ya<<endl;

cout << "Enter private key for Bob (Xb): ";
cin>>Xb;
cout << endl;
```

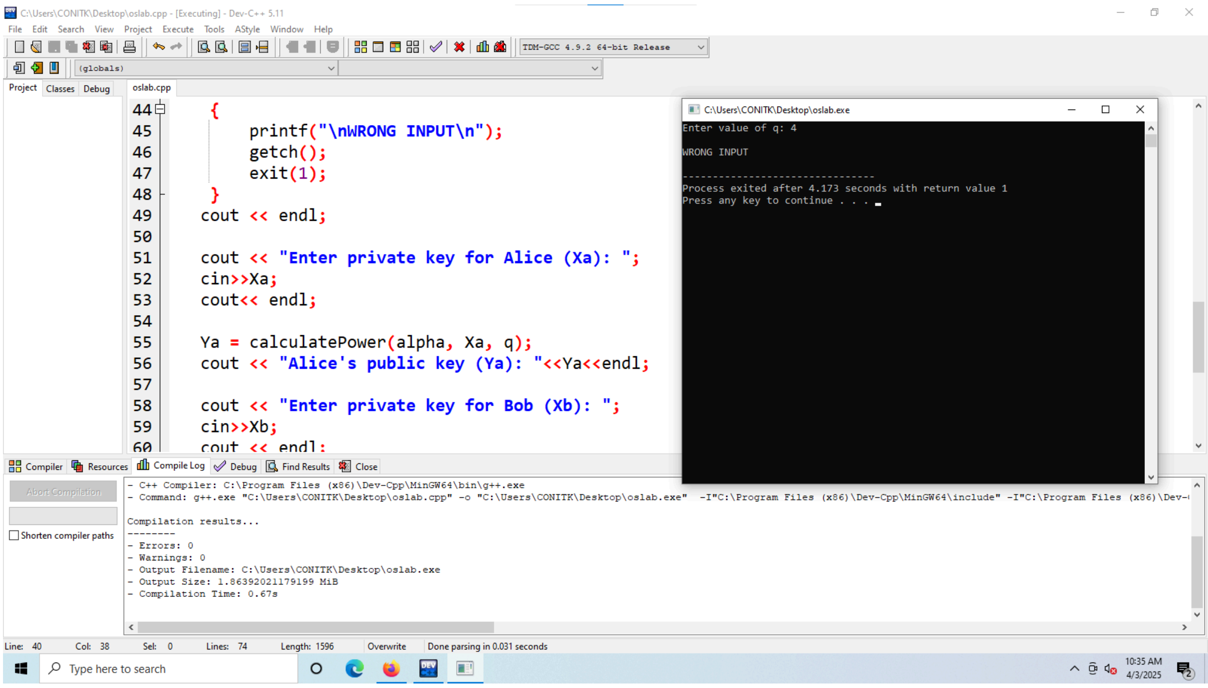
```
    Yb = calculatePower(alpha, Xb, q);
    cout << "Bob's public Key (Yb): "<<Yb<<endl;

    Ka = calculatePower(Yb, Xa,q);
    Kb = calculatePower(Ya, Xb,q);

    cout << "Secret key for Alice: " << Ka << endl;
    cout << "Secret key for Bob: " << Kb << endl;
    cout<<endl<<endl;
    cout<<"The Key for both Alice and Bob is same"<<endl;
    return 0;
}
```

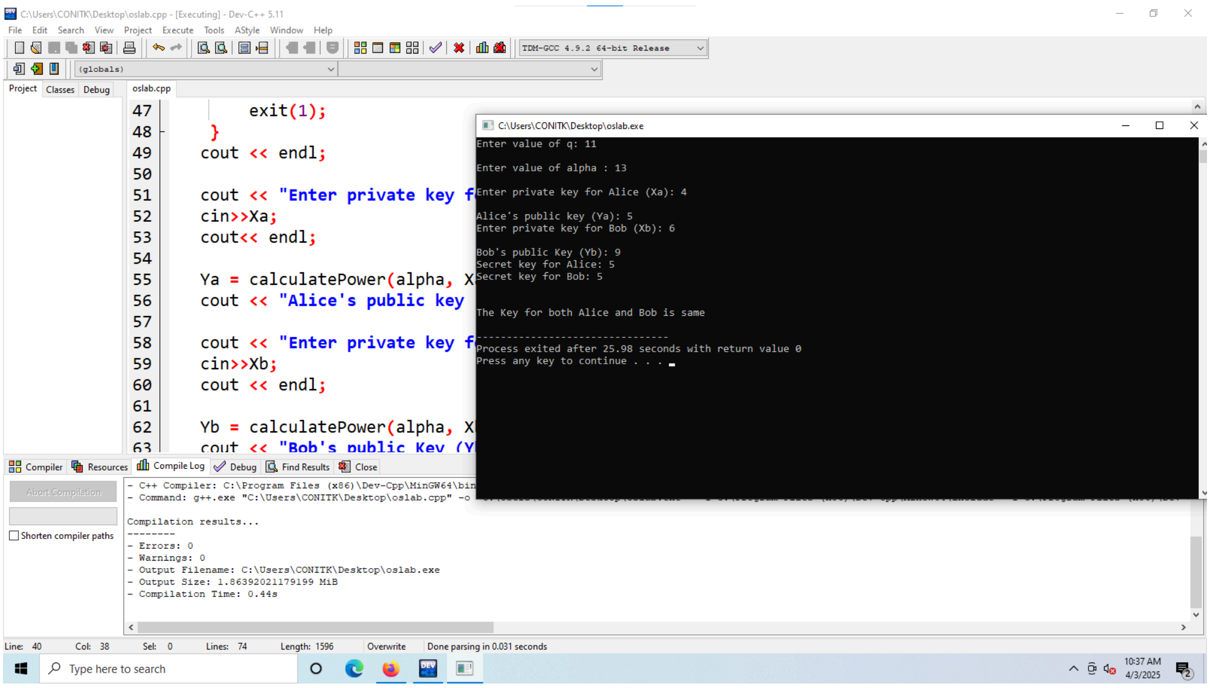
OUTPUT :

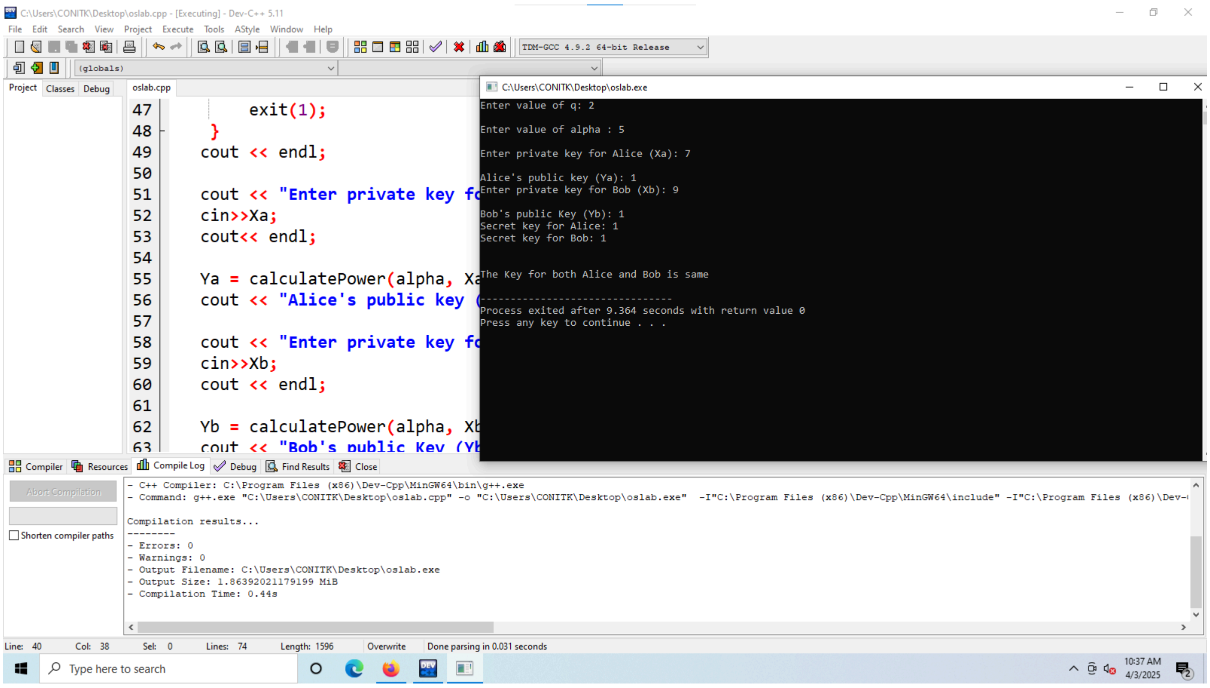

C:\Users\CONITK\Desktop\oslab.exe
Enter value of q: 7
Enter value of alpha : 11
Enter private key for Alice (Xa): 3
Alice's public key (Ya): 1
Enter private key for Bob (Xb): 7
Bob's public Key (Yb): 4
Secret key for Alice: 1
Secret key for Bob: 1
The Key for both Alice and Bob is same

Process exited after 9.688 seconds with return value 0
Press any key to continue . . .
Compilation results...
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\CONITK\Desktop\oslab.exe
- Output Size: 1.86392021179199 MiB
- Compilation Time: 0.44s


C:\Users\CONITK\Desktop\oslab.exe
Enter value of q: 2
Enter value of alpha : 4
WRONG INPUT
Compilation results...
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\CONITK\Desktop\oslab.exe
- Output Size: 1.86392021179199 MiB
- Compilation Time: 0.44s


C:\Users\CONITK\Desktop\oslab.cpp - [Executing] - Dev-C++ 5.11
{
printf("\nWRONG INPUT\n");
getch();
exit(1);
}
cout << endl;
cout << "Enter private key for Alice (Xa): ";
cin>>Xa;
cout<< endl;
Ya = calculatePower(alpha, Xa, q);
cout << "Alice's public key (Ya): "<<Ya<<endl;
cout << "Enter private key for Bob (Xb): ";
cin>>Xb;
cout << endl;
C:\Users\CONITK\Desktop\oslab.exe
Enter value of q: 4
WRONG INPUT
Process exited after 4.173 seconds with return value 1
Press any key to continue . . .
Compilation results...
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\CONITK\Desktop\oslab.exe
- Output Size: 1.86392021179199 MiB
- Compilation Time: 0.67s
10:35 AM
4/3/2025


C:\Users\CONITK\Desktop\oslab.cpp - [Executing] - Dev-C++ 5.11
C:\Users\CONITK\Desktop\oslab.exe
Enter value of q: 11
Enter value of alpha : 13
Enter private key for Alice (Xa): 4
Alice's public key (Ya): 5
Enter private key for Bob (Xb): 6
Bob's public Key (Yb): 9
Secret key for Alice: 5
Secret key for Bob: 5
The Key for both Alice and Bob is same
Process exited after 25.98 seconds with return value 0
Press any key to continue . . .
Compilation results...
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\CONITK\Desktop\oslab.exe
- Output Size: 1.86392021179199 MiB
- Compilation Time: 0.44s
10:37 AM
4/3/2025


C:\Users\CONITK\Desktop\oslab.cpp - [Executing] - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 4.9.2 64-bit Release
(globals)
Project Classes Debug
oslab.cpp
exit(1);
}
cout << endl;
cout << "Enter private key fo
cin>>Xa;
cout<< endl;
Ya = calculatePower(alpha, Xa
cout << "Alice's public key (
cout << "Enter private key fo
cin>>Xb;
cout << endl;
Yb = calculatePower(alpha, Xb
C:\Users\CONITK\Desktop\oslab.exe
Enter value of q: 2
Enter value of alpha : 5
Enter private key for Alice (Xa): 7
Alice's public key (Ya): 1
Enter private key for Bob (Xb): 9
Bob's public Key (Yb): 1
Secret key for Alice: 1
Secret key for Bob: 1
The Key for both Alice and Bob is same

Process exited after 9.364 seconds with return value 0
Press any key to continue . . .
Compiler Resources Compile Log Debug Find Results Close
Abort Compilation
Shorten compiler paths
- C++ Compiler: C:\Program Files (x86)\Dev-Cpp\MinGW64\bin\g++.exe
- Command: g++.exe "C:\Users\CONITK\Desktop\oslab.cpp" -o "C:\Users\CONITK\Desktop\oslab.exe" -I"C:\Program Files (x86)\Dev-Cpp\MinGW64\include" -I"C:\Program Files (x86)\Dev-
Compilation results...

- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\CONITK\Desktop\oslab.exe
- Output Size: 1.86392021179199 MiB
- Compilation Time: 0.44s
Line: 40 Col: 38 Sel: 0 Lines: 74 Length: 1596 Overwrite Done parsing in 0.031 seconds
Type here to search
10:37 AM
4/3/2025